

Product Requirements Document

Marine Oil Spill Detection & AIS-Based Vessel Attribution System

SIH Problem Statement — NTRO — H3-Indexed Pipeline

Version: 1.0

Team size: 6

Build window: 3 days (hackathon)

Document purpose: This PRD is the single source of truth for the system. It is written to be consumable by both human team members and AI coding agents (e.g. Claude Code) as grounding context — every module, data contract, and algorithm is specified explicitly enough that an agent can implement it without needing to infer intent.

1. Problem Statement Recap

Routine, deliberate operational discharge of oil (tank washings, bilge water, oily ballast) from vessels at sea is illegal under MARPOL Annex I but almost never prosecuted, because by the time a slick is observed, the responsible vessel is long gone and no evidence chain links slick to ship.

The system must do three things:

1. **Detect & characterise** — find oil slicks in satellite SAR imagery, distinguish them from visual look-alikes (calm water, algae, wind shadows), and describe their geometry.
2. **Hindcast & forecast** — model ocean physics to determine where the slick *came from* (for attribution/forensics) and where it is *going* (for spill response).
3. **Attribute** — reconstruct vessel traffic (AIS) in the backtracked origin region and time window, and produce a ranked, explainable list of suspect vessels — including flagging vessels that were physically present (visible on radar) but not broadcasting AIS ("dark vessels").

The deliverable is not a cleanup tool. It is a **deterrence and evidence-generation tool**. The core value proposition: "*We make illegal discharge a risky decision to make,*" not "*we help clean up oil after the fact.*"

2. Design Decision: H3 Hexagonal Discretization

2.1 Why this changes the architecture

The original/naive approach treats the slick as a continuous polygon and the backtracked drift field as a continuous probability surface, then performs point-in-polygon and spatial-join operations against AIS tracks. This is computationally heavier and produces fuzzy, hard-to-audit outputs.

This system instead converts every spatial object — slick polygon, drift particle cloud, AIS vessel positions — into **H3 discrete global grid cells**. This changes the core attribution operation from a geometric spatial join into a **hash/set lookup**, which is faster to implement correctly under time pressure and produces a **discrete, reproducible, auditable** evidence artifact (a list of cell IDs) rather than a floating-point polygon that's harder to defend as evidence.

2.2 H3 resolution choice

- **Resolution 7** (~5.16 km² per cell, ~1.4 km edge length) is the default for open-ocean backtracking.
- **Resolution 8** (~0.74 km² per cell, ~0.53 km edge length) may be used near-coast if tighter precision is demonstrably needed.

- **Justification to state explicitly in any report/pitch:** HYCOM ocean current data has native resolution on the order of ~8 km; resolving hex cells finer than the input physics data creates false precision. Resolution 7 is the defensible choice and should not be second-guessed mid-build.
- The resolution **MUST** be fixed once, recorded in the case file schema (see §7.4), and never silently changed between pipeline runs — reproducibility depends on this.

2.3 What H3 does NOT solve

H3 discretization does not remove the need for a proper drift model (OpenDrift), a proper look-alike filter, or a proper scoring model. It only changes *how* spatial matching is implemented. Do not let this framing shrink the scope of Stages 1, 4, or 5 below — attribution quality still depends entirely on drift-model correctness and scoring-feature quality, not on the indexing scheme.

3. System Architecture

3.1 High-level pipeline (five stages, strictly sequential data dependency)

```
[SAR Scene]
-> Stage 1: Detection & Look-Alike Filtering
-> [Slick Polygon + Confidence]
-> Stage 2: H3 Indexing & Lagrangian Backtracking
-> [Time-indexed Origin Hex Corridor]
-> Stage 3: AIS Ingestion & Spatiotemporal Filtering
-> [Candidate Vessel Set]
-> Stage 4: Multi-Factor Scoring & Dark Vessel Detection
-> [Ranked Suspect List + Dark Vessel Flags]
-> Stage 5: Dashboard & Case File Generation
-> [Interactive UI + Exportable Evidence Package]
```

Each stage's output is a well-defined data contract (see §7) that the next stage consumes. This is deliberate: it allows different team members to build stages in parallel against a mocked/stubbed contract on Day 1, then wire them together on Day 2–3 without waiting on each other.

3.2 Technology stack

Layer	Technology	Notes
SAR data source	Sentinel-1 GRD via Google Earth Engine	No local download needed; fastest path for hackathon. Fallback: ASF DAAC / Copernicus Data Space Ecosystem
SAR preprocessing	GEE built-in calibration, or SNAP/snappy/pyroSAR if working locally	Calibration → speckle filter (Refined Lee) → terrain correction
Segmentation model	U-Net or DeepLabv3+ (PyTorch), pretrained backbone, fine-tuned	Dataset: Krestenitis et al. labelled SAR oil-spill dataset (includes oil spill, look-alike, ship, land, sea classes)
Look-alike filter	Rule-based: wind speed gate + damping ratio + shape features	ERA5 reanalysis for wind
Geometric analysis	rasterio , shapely , geopandas	Area, perimeter, major/minor axis, eccentricity, centroid
H3 indexing	h3-py	Resolution 7 (default)
Ocean currents	HYCOM (OPeNDAP, free) or CMEMS/Copernicus Marine	Backward-in-time integration input

Layer	Technology	Notes
Wind data	ERA5 (Copernicus Climate Data Store)	Used both for look-alike filtering and drift wind-drift-factor
Drift model	OpenDrift / OpenOil (MET Norway)	Negative timestep = backward integration. This is the single highest-leverage library in the stack
AIS data	MarineCadastre.gov (US, free bulk) / Global Fishing Watch API / synthetic AIS (explicitly permitted by the PS for Indian waters)	Historical positional data, ingested and hex-indexed
Ship detection (SAR)	CFAR (Constant False Alarm Rate) classical detector	Alternative: small YOLO fine-tuned on SSDD/HRSID/LS-SSDD datasets
Scoring model	Logistic regression or gradient boosting (interpretable)	Explicitly NOT a deep net — evidence requires explainability
Frontend/UI	MapLibre GL JS / Deck.gl / Kepler.gl, React or Streamlit	Time-slider driven map
Case file export	GeoJSON + PDF generation	Hashed inputs for tamper-evidence
Secrets	Environment variables / secrets manager	Copernicus, CMEMS, GFW all require credentials — never commit keys

4. Stage-by-Stage Functional Specification

Stage 1 — Satellite Ingestion & Slick Detection

Input: Area of interest (bounding box or region name), target date/time window.

Steps:

- Query Sentinel-1 GRD scenes over the AOI/time window via Google Earth Engine.
- Apply radiometric calibration, speckle filtering (Refined Lee), and terrain correction.
- Run the segmentation model (U-Net/DeepLabv3+) over the calibrated scene to produce a binary/multi-class dark-pixel mask (classes: oil spill, look-alike, ship, land, sea surface — per Krestenitis taxonomy).
- Vectorize the oil-spill class mask into polygon(s) using `rasterio` / `shapely` .
- Look-alike filter (critical gate — do not skip or stub this):**
 - Pull ERA5 wind speed for the scene's time and location.
 - Reject** the candidate region if wind speed < 2 m/s or > 12 m/s (oil is undetectable in SAR outside this band — below it everything looks dark/glassy, above it the damping signature washes out).
 - Compute **damping ratio**: mean backscatter inside the dark region vs. local sea background. Mineral oil has a distinctive damping signature; natural look-alikes typically do not.
 - Compute **shape features**: eccentricity, solidity, perimeter²/area (complexity). Operational discharges are typically long/thin/linear (a moving vessel leaves a line); natural calm patches are typically broad and amorphous.
 - Compute **alignment**: does the polygon's major axis align with a plausible vessel heading / known shipping lane? A linear slick parallel to a shipping lane is a signature, not coincidence.
- Output the **confirmed** slick polygon(s) with a per-polygon confidence score and the full feature vector used to confirm it (for auditability — this becomes part of the case file).

Output contract: see §7.1.

Explicit requirement: The demo MUST include at least one instance where a low-wind dark patch is correctly rejected by this filter, shown live. This is the single most persuasive thirty seconds in the entire pipeline for a judge/evaluator audience — do not treat it as optional polish.

Stage 2 — Hexagonal Discretization & Physical Backtracking

Input: Confirmed slick polygon(s) from Stage 1, scene acquisition timestamp.

Steps:

1. **H3 indexing of the slick:** Convert the confirmed polygon into the set of H3 cells at the chosen resolution that it covers. Call this set `H_0` (the origin/observation-time hex set).
2. **Particle seeding:** Seed one virtual Lagrangian particle at the centroid of each cell in `H_0` (alternative: seed multiple particles per cell for a denser ensemble if compute allows — note as a stretch goal, not baseline).
3. **Backward integration (OpenDrift):** Run OpenDrift/OpenOil with a **negative timestep**, driven by:
 - o HYCOM ocean surface currents (primary advection)
 - o ERA5 surface winds, applied with a wind-drift factor of approximately 3% of wind speed (standard oil-drift parameterization), plus stochastic horizontal diffusion to represent turbulent uncertainty.
 - o Integrate backward at minimum three checkpoints: **-6h, -12h, -24h** from the scene acquisition time. (Extending further back is possible but uncertainty grows quickly — see §8 Known Limitations.)
4. **Re-binning to hexagons:** At each backward timestep, take the current particle cloud positions and convert them back into the set of H3 cells they occupy. This produces a **time-indexed corridor**:

```
Corridor = {
  t0 (observation time): H_0,
  t_minus_6h: H_{-6},
  t_minus_12h: H_{-12},
  t_minus_24h: H_{-24}
}
```

5. Each hex cell in each timestep's set should retain a **particle density** (count of particles that landed in it) — this becomes an implicit probability weight, used later for scoring, without needing to build a full continuous probability surface.

Output contract: see §7.2.

Framing note for the report/pitch: Backtracking uncertainty compounds the further back you integrate. A 24-hour backtrack produces a spread-out corridor of hexagons, not a single point. This is correct physics, not a flaw — present it as an uncertainty-aware origin region. A system that claims a single pinpoint origin after 24 hours of backward drift is either wrong or lying; say this explicitly if asked.

Stage 3 — AIS Vessel Ingestion & Spatiotemporal Filtering

Input: Historical AIS data for the region and time window spanned by the Stage 2 corridor; the corridor itself.

Steps:

1. **Ingest AIS:** Pull historical AIS logs (MarineCadastre.gov for demo regions with free coverage, Global Fishing Watch API, or explicitly-labelled **synthetic AIS** for Indian-waters demonstration — the problem statement explicitly permits synthetic AIS given real Indian coastal AIS is restricted to DG Shipping/ICG and will not be released to a hackathon team).
2. **Standardize:** Snap every AIS position report to 5-minute timestamp intervals (interpolate if a vessel's native reporting interval is coarser or finer).
3. **Hex-index every vessel position:** For every `(vessel_id, lat, lon, timestamp)` tuple, compute the corresponding H3 cell at the pipeline's fixed resolution, producing `(vessel_id, hex_id, timestamp)`.

4. **Corridor query (primary match):** For each timestep in the Stage 2 corridor, perform a set-intersection / hash lookup: does any `(hex_id, timestamp)` from the AIS index match a hex in that timestep's corridor set `H_{-t}` ? This is an $O(1)$ dictionary lookup per AIS record, not a spatial join — this is the entire point of the H3 approach.
5. **Bounded k-ring expansion (fallback, only if primary match returns few/no candidates):** Expand each corridor hex by 1–2 neighboring rings (`k=1` , `k=2` using H3's k-ring function) and re-query. Each expansion ring must carry an explicit **confidence-decay penalty** (e.g. weight $1/(1+k)$) so that a vessel found only via `k=2` expansion is never scored as strongly as one found in the primary corridor. This prevents falsely implicating distant, uninvolved vessels while still surfacing near-miss candidates for a human reviewer.
6. Output the **candidate vessel set**: every vessel with at least one corridor or k-ring match, tagged with which hex/timestep/ring it matched on.

Output contract: see §7.3.

Stage 4 — Multi-Factor Scoring & Dark Vessel Detection

Input: Candidate vessel set from Stage 3; original SAR scene from Stage 1.

Part A — Probabilistic/explainable scoring of AIS-visible candidates:

For each candidate vessel, compute an explainable score as a weighted combination of at minimum:

Feature	Description	Why it matters
Corridor overlap	Duration/weight of time the vessel's track intersected the backtracked corridor (using particle-density weights from Stage 2 as the probability mass)	Core likelihood signal — replaces naive "nearest vessel" proximity reasoning
Heading alignment	Angle between vessel heading and the slick's major axis orientation	A vessel discharging while underway leaves a slick aligned with its direction of travel
Speed anomaly	Detected speed drops consistent with a "pumping" operation (slow, steady speed rather than normal transit speed)	Behavioral signature of an active discharge event
AIS gap history	Whether this vessel has a history of AIS transponder gaps (especially near this timeframe)	A vessel that goes dark repeatedly, especially right before/during the estimated discharge window, is more suspicious than one with a clean history

Use an **interpretable model** (logistic regression or gradient boosting with feature importances) — explicitly do NOT use a black-box deep model for this step. The output must be explainable per-feature, because the entire point of this stage is producing evidence, not just a score.

Formally, the corridor-overlap term can be expressed as a likelihood integral over the discretized probability field:

$$\text{Score_corridor}(\text{vessel}) = \sum_t \sum_{h \in \text{corridor}(t)} [\text{particle_density}(h,t) \times 1(\text{vessel at hex } h, \text{ time } t)] \times \text{decay_weight}(k\text{-ring})$$

The full score is a weighted sum of this term plus the heading, speed-anomaly, and AIS-gap-history terms, with weights that must be stated explicitly and included in the case file (see §7.4) — this is what makes the score auditable rather than a mystery number.

Part B — Dark vessel detection (independent, parallel check):

- Run a **CFAR (Constant False Alarm Rate)** detector over the raw SAR scene to find all physical metal-hull reflections (ships), independent of any AIS data.
- Cross-reference every CFAR-detected ship position against the AIS candidate set from Stage 3.
- Classify each CFAR detection:

SAR ship detected?	AIS track present at that location/time?	Interpretation
✓	✓	Identified vessel — proceed to Part A scoring
✓	✗	DARK VESSEL — physically present, not broadcasting. Flag as highest-priority evidence, separate from and above the ranked suspect list.
✗	✓	AIS-only signal — vessel may be outside SAR scene extent, or position may be spoofed. Note but do not over-weight.

A CFAR-detected ship near the head/origin of the corridor with no AIS match is the single strongest piece of evidence the system can produce, and it comes essentially free from imagery already being processed in Stage 1. It must be surfaced distinctly in the UI (§Stage 5), not buried inside the general ranked list.

Explicit null-result requirement: If no candidate scores above a defined confidence threshold, the system must output "**no candidate identified**" rather than force-ranking a weak match to the top of a list. This should be explicitly labelled as a probable dark-vessel-outside-SAR-coverage event or an out-of-AIS-coverage event. A system that always names a suspect will eventually name an innocent one — restraint here is a designed behavior, not a gap.

Output contract: see §7.4 (feeds directly into the case file).

Stage 5 — Interactive Dashboard & Case File Generation

Functional requirements:

- Geospatial map UI**, layered:
 - Base map
 - Raw/processed SAR scene as a raster overlay
 - Detected slick polygon
 - Animated H3 hex corridor (color/opacity indicating particle density per cell, per timestep)
 - AIS vessel tracks (all candidates; dark-vessel flags visually distinct — e.g. red/pulsing marker)
 - Time slider** driving the animation from scene-observation-time backward through -6h/-12h/-24h, and optionally forward for the spill-response forecast direction
- Suspect ranking panel:** ordered list of candidate vessels with total score and a visible per-feature breakdown (corridor overlap, heading alignment, speed anomaly, AIS-gap history) — this must be a real breakdown, not just a bar chart of the total.
- Dark vessel alert panel:** separate, visually prioritized from the ranked list, showing any CFAR-detected/no-AIS-match vessels.
- Auditable evidence export:** a downloadable case file (GeoJSON + PDF) containing:
 - SAR scene ID and acquisition timestamp
 - Processing parameters (segmentation model version, look-alike filter thresholds used)
 - H3 resolution used
 - Drift model configuration (currents source, wind source, wind-drift factor, diffusion parameters, backward timesteps run)
 - The full origin hex corridor (as a list of (hex_id, timestep) pairs)
 - AIS query bounds (spatial + temporal)
 - Every candidate vessel's full feature vector and score, with weights used
 - A hash of the input data set, for tamper-evidence

Output contract: see §7.5.

5. Non-Functional Requirements

Requirement	Detail
Offline-capable demo	Pre-cache SAR scene(s), currents, wind, and AIS data for the chosen demo scenario before presenting. Do not depend on live API calls or venue wifi during the actual demo.
Reproducibility	Given the same cached inputs and the same recorded parameters (resolution, thresholds, model version), running the pipeline twice must produce the same corridor and the same suspect list. This is required for the "evidence-grade" framing to hold up under questioning.
Explainability	No black-box scoring. Every number shown to the user in the suspect panel must be traceable to a named feature and weight.
Secrets handling	All API credentials (Copernicus, CMEMS, GFW) in environment variables / a secrets manager. Never committed to the repository. If a key is accidentally exposed, rotate it immediately.
Honest uncertainty	Backward-drift outputs are corridors/regions, not points. Do not present false precision anywhere in the UI or case file.
Graceful null result	The system must be able to output "no suspect identified" without being forced to name one.

6. Explicit Non-Goals (Scope Boundaries for the 3-Day Build)

To keep six people from overbuilding in three days, the following are explicitly OUT of scope for the hackathon prototype:

- Real, restricted Indian Coast Guard/DG Shipping AIS data — use synthetic AIS or a free-coverage region (US/EU) for the working demo, and describe the Indian-waters deployment path in the report as future work.
- A statistically validated attribution accuracy figure — there is no public ground-truth dataset of "slick X was caused by vessel Y, confirmed," so no such metric should be claimed. Frame the system as a ranked-suspect / probabilistic-scoring tool validated against constructed test scenarios, not as a system with a reported accuracy percentage.
- Live/streaming ingestion during the demo — everything is pre-cached (see §5).
- Blockchain/hash-chaining beyond a single input hash for tamper-evidence, unless someone actually has time to build it properly — do not bolt this on for buzzword value.
- Age-estimation of the slick beyond a best-effort attempt using spreading physics + backtrack consistency, explicitly caveated as approximate — the source problem statement itself flags this as "if feasible," not a hard requirement.

7. Data Contracts (Inter-Stage Interfaces)

These schemas are the actual integration contract between team members' modules. Build against these from Day 1 so stages can be developed in parallel and wired together later without renegotiating formats mid-build.

7.1 Stage 1 Output — Confirmed Slick

```
{
  "scene_id": "string",
  "acquisition_time": "ISO8601 timestamp",
  "polygons": [
    {
      "polygon_id": "string",
      "geometry": "GeoJSON Polygon",
```

```

"confidence": "float 0-1",
"geometry_features": {
  "area_km2": "float",
  "perimeter_km": "float",
  "major_axis_km": "float",
  "minor_axis_km": "float",
  "eccentricity": "float",
  "orientation_deg": "float"
},
"lookalike_filter": {
  "wind_speed_ms": "float",
  "wind_gate_passed": "boolean",
  "damping_ratio": "float",
  "shape_gate_passed": "boolean",
  "final_decision": "confirmed | rejected",
  "rejection_reason": "string | null"
}
}
]
}

```

7.2 Stage 2 Output — Origin Hex Corridor

```

{
  "h3_resolution": "integer (7 default)",
  "scene_id": "string",
  "corridor": {
    "t0": { "hex_ids": ["string"], "particle_density": {"hex_id": "int"} },
    "t_minus_6h": { "hex_ids": ["string"], "particle_density": {"hex_id": "int"} },
    "t_minus_12h": { "hex_ids": ["string"], "particle_density": {"hex_id": "int"} },
    "t_minus_24h": { "hex_ids": ["string"], "particle_density": {"hex_id": "int"} }
  },
  "drift_config": {
    "currents_source": "HYCOM",
    "wind_source": "ERA5",
    "wind_drift_factor": 0.03,
    "diffusion_coefficient": "float",
    "particle_count": "int"
  }
}

```

7.3 Stage 3 Output — Candidate Vessel Set

```

{
  "candidates": [
    {
      "vessel_id": "MMSI or synthetic ID",
      "matches": [
        {
          "hex_id": "string",
          "timestep": "t0 | t_minus_6h | t_minus_12h | t_minus_24h",
          "match_type": "primary | k_ring",
          "k_ring": "int (0 for primary)",
          "decay_weight": "float"
        }
      ]
    }
  ],
  "ais_query_bounds": {
    "spatial": "GeoJSON bounding polygon",
    "temporal": {"start": "ISO8601", "end": "ISO8601"}
  }
}

```



```
}  
}
```

7.4 Stage 4 Output — Scored Suspects + Dark Vessels (feeds Case File)

```
{  
  "ranked_suspects": [  
    {  
      "vessel_id": "string",  
      "total_score": "float",  
      "feature_breakdown": {  
        "corridor_overlap_score": "float",  
        "heading_alignment_score": "float",  
        "speed_anomaly_score": "float",  
        "ais_gap_history_score": "float"  
      },  
      "weights_used": {  
        "corridor_overlap": "float",  
        "heading_alignment": "float",  
        "speed_anomaly": "float",  
        "ais_gap_history": "float"  
      }  
    }  
  ],  
  "dark_vessels": [  
    {  
      "cfar_detection_id": "string",  
      "position": "GeoJSON Point",  
      "timestamp": "ISO8601",  
      "ais_match_found": false,  
      "proximity_to_corridor": "string (hex_id or null)"  
    }  
  ],  
  "null_result": "boolean – true if no suspect exceeded confidence threshold"  
}
```

7.5 Stage 5 Output — Case File

```
{  
  "case_id": "string",  
  "generated_at": "ISO8601",  
  "scene_id": "string",  
  "h3_resolution": "integer",  
  "processing_parameters": {  
    "segmentation_model_version": "string",  
    "lookalike_thresholds": {"wind_min_ms": 2, "wind_max_ms": 12},  
    "drift_config": "as in 7.2",  
    "scoring_weights": "as in 7.4"  
  },  
  "corridor": "as in 7.2",  
  "ais_query_bounds": "as in 7.3",  
  "ranked_suspects": "as in 7.4",  
  "dark_vessels": "as in 7.4",  
  "input_data_hash": "string (sha256)"  
}
```

8. Known Limitations (state these proactively, do not let a judge surface them first)

- **Backward-drift uncertainty compounds with time.** A 24-hour backtrack produces a wide, multi-hex corridor, not a point. This is correct behavior, not a bug — present it as an uncertainty region.
- **No public attribution ground-truth dataset exists.** No accuracy percentage should be reported for "correctly identified the guilty vessel." Validate instead against constructed test scenarios with known synthetic answers, and say so explicitly.
- **Sentinel-1 revisit frequency is not continuous.** A slick can fully form and dissipate between satellite passes — this system detects what is imaged, not everything that occurs.
- **AIS is cooperative, not enforced.** A vessel intending to discharge illegally has an incentive to go dark before doing so — this is precisely why the dark-vessel detection path (Stage 4, Part B) exists and should be emphasized as the system's answer to this limitation, not treated as a separate feature.
- **Real Indian coastal AIS is restricted** to DG Shipping / Indian Coast Guard and is not obtainable for a hackathon prototype. The problem statement explicitly permits synthetic AIS for this reason.

9. Work Breakdown — 6-Person Team, 3-Day Build

Day 1 — Foundations (parallel tracks; sync and integrate contracts at end of day against §7 schemas)

Owner	Track	Deliverable	Data contract owned
P1	SAR ingestion + preprocessing	GEE pipeline pulling and calibrating a chosen demo-event Sentinel-1 scene	Feeds into §7.1
P2	Segmentation model	U-Net/DeepLabv3+ fine-tuned on Krestenitis dataset; outputs raw slick mask/polygon	§7.1 (geometry)
P3	Look-alike filter	ERA5 wind pull, wind gate, damping ratio, shape-feature computation	§7.1 (lookalike_filter block)
P4	H3 pipeline core	Polygon → H3 conversion utilities, k-ring expansion utilities, corridor data structure implementation	§7.2 skeleton
P5	AIS ingestion	Pull or synthesize AIS for the demo region; snap-to-5-min-interval; hex-index all vessel positions	§7.3 skeleton
P6	UI shell + repo scaffolding	Map base (MapLibre/Deck.gl), project structure, environment/secrets setup, CI-safe <code>.gitignore</code> for credentials	N/A (infra)

Day 2 — Core Logic + Fusion

Owner	Track	Deliverable
P1 + P2	OpenDrift integration	Backward particle ensemble seeded from Stage 1 polygon, run at -6h/-12h/-24h, re-binned into H3 corridor per §7.2
P3	Scoring model	Feature engineering for corridor overlap, heading alignment, speed anomaly, AIS-gap history; logistic regression/GBM scorer with explicit weights
P4	Corridor ↔ AIS matching	Wire H3 corridor output (P1+P2) to AIS hex index (P5); implement k-ring expansion with decay weighting per §Stage 3
P5	CFAR ship detection	CFAR detector on SAR scene; cross-reference against AIS-matched vessel set to produce dark-vessel candidate list
P6	UI: timeline + layers	Time slider wired to corridor animation; AIS track overlay; SAR raster layer

Owner	Track	Deliverable
P1 + P4	End-to-end pipeline wiring	Single orchestrated pipeline: scene → confirmed slick → corridor → candidates → scored suspects + dark vessels, per §7 contracts throughout
P2 + P3	Suspect ranking panel + explainability UI	Per-feature score breakdown display; explicit "no candidate identified" null-result state implemented and demoed
P5	Case file export	GeoJSON/PDF generator implementing the §7.5 schema, including input-data hash
P6	Demo scenario + offline caching	Pre-cache all data for the chosen demo scenario (§5 requirement); rehearse the look-alike-rejection moment and the dark-vessel-flag moment specifically
All	Integration testing + rehearsal	Full dry run against cached data; fix cross-module contract mismatches; time the pitch

Non-negotiable demo checklist (Day 3, end of day)

- ☐ One complete real (or realistic synthetic) spill event, start to finish
 - ☐ A visible, live look-alike rejection (low-wind patch correctly discarded)
 - ☐ An animated backward hex corridor converging toward a plausible origin
 - ☐ At least one dark-vessel flag (CFAR detection with no AIS match)
 - ☐ Ranked suspect list with visible per-feature score breakdown
 - ☐ A "no candidate identified" null-result path demonstrated (even if only in a second scenario)
 - ☐ Exportable case file downloadable from the UI
-

10. Positioning for Pitch / Report

- Lead with **deterrence, not cleanup**: "We make illegal discharge a risky decision" is the core claim.
 - The **moat** is not detection (well-solved in literature) — it is the combination of (1) physics-based probabilistic backtracking instead of naive proximity matching, (2) dark-vessel fusion from CFAR+AIS cross-referencing, and (3) rigorous look-alike rejection. All three are buildable in three days and together they are what existing operational systems (e.g. EMSA CleanSeaNet) are comparatively weak on.
 - Cite **SkyTruth Cerulean** as the closest open-source reference implementation and differentiate explicitly: this system adds physics-based backward corridor attribution (not just nearest-AIS-track), dark-vessel fusion, and an H3-indexed evidence structure.
 - Cite **Paolo et al., Nature 2024** (SAR+AIS fusion for global industrial vessel activity) as the grounding citation for the dark-vessel argument.
 - Cite **EMSA CleanSeaNet** as proof the overall architecture works at operational/regional scale — position this system as the India-context version with a stronger, physics-grounded attribution layer.
-

End of document.